

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B

COURSE OUTCOME COVERED

***CO4:** Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

QUESTIONS: 5 Total Marks: 50 Duration: 60 Minutes Annexure A

Comparative Analysis

Code of Conduct:

I A.Shiva Shankar Reddy(192425174) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Shiva Shankar Reddy

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object	Stereo Vision	Structure from Motion
A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A = 2, B = 3, C = 4

(a) Compute absolute error for both methods.

Formula: Absolute Error = $|\text{Estimated Value} - \text{Actual Value}|$

Stereo Vision:

$$\text{A: } |2.0 - 2| = 0.0$$

$$\text{B: } |3.0 - 3| = 0.0$$

$$\text{C: } |4.0 - 4| = 0.0$$

$$\text{Average Absolute Error (Stereo)} = (0.0 + 0.0 + 0.0) / 3 = 0.0 \text{ m}$$

Structure from Motion (SfM):

$$\text{A: } |2.5 - 2| = 0.5$$

$$\text{B: } |3.8 - 3| = 0.8$$

$$\text{C: } |5.0 - 4| = 1.0$$

$$\text{Average Absolute Error (SfM)} = (0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = 0.767 \text{ m}$$

(b) Compare and evaluate which method is more accurate.

Stereo Vision has an average absolute error of 0.0 m, while Structure from Motion has an average absolute error of 0.767 m. Since a lower absolute error indicates higher accuracy, Stereo Vision is more accurate than Structure from Motion for this dataset. SfM consistently overestimates depth, with the error increasing as depth increases, suggesting scale drift, which is a known limitation of monocular SfM without proper calibration.

2. Pattern Recognition Accuracy Comparison

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Compute accuracy for both systems.

Formula: $\text{Accuracy} = (\text{Correct Predictions} / \text{Total Samples}) \times 100$

Traditional System: $\text{Accuracy} = (85 / 100) \times 100 = 85\%$

Deep Learning System: $\text{Accuracy} = (92 / 100) \times 100 = 92\%$

(b) Compare and evaluate which system performs better.

The Deep Learning system achieves an accuracy of 92%, which is 7 percentage points higher than the Traditional system's accuracy of 85%. Therefore, the Deep Learning system performs better at pattern recognition on this dataset, indicating it generalizes more effectively to the given samples.

3. Precision-Recall Trade-off Analysis

Model	TP	FP	FN
A	70	30	20
B	85	25	25

(a) Compute precision and recall for both models.

Formulas: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$ $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Model A:

$\text{Precision} = 70 / (70 + 30) = 70 / 100 = 0.70$ (70%)

$\text{Recall} = 70 / (70 + 20) = 70 / 90 = 0.778$ (77.8%)

Model B:

$\text{Precision} = 85 / (85 + 25) = 85 / 110 = 0.773$ (77.3%)

$\text{Recall} = 85 / (85 + 25) = 85 / 110 = 0.773$ (77.3%)

(b) Compare and evaluate which model is more balanced.

Model A has Precision = 0.70 and Recall = 0.778, a gap of 0.078 between the two metrics. Model B has Precision = 0.773 and Recall = 0.773, showing no gap at all. Since a balanced model has precision and recall values that are close to each other, Model B is more balanced than Model A. Model A favors recall over precision, meaning it misses fewer positives but produces more false positives, while Model B maintains an even trade-off between the two.

4. F1-Score Comparison

Model	Precision	Recall
A	0.8	0.6
B	0.75	0.75

(a) Compute F1-score for both models.

Formula: $F1\text{-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Model A: $F1 = 2 \times (0.8 \times 0.6) / (0.8 + 0.6) = 2 \times 0.48 / 1.4 = 0.96 / 1.4 = 0.686$

Model B: $F1 = 2 \times (0.75 \times 0.75) / (0.75 + 0.75) = 2 \times 0.5625 / 1.5 = 1.125 / 1.5 = 0.750$

(b) Compare and evaluate which model is better.

Model B has an F1-score of 0.750, higher than Model A's F1-score of 0.686. Since the F1-score reflects the harmonic mean of precision and recall, Model B provides a better overall balance between the two metrics and is therefore considered the better-performing model, despite Model A having a higher precision value alone.

5. Optical Flow Consistency

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

Formula: $\text{Average Motion} = \text{Sum of Motion Values} / \text{Number of Frames}$

Method A: $(4 + 5 + 6) / 3 = 15 / 3 = 5.0$

Method B: $(6 + 8 + 9) / 3 = 23 / 3 = 7.667$

(b) Compare consistency and evaluate which is more stable.

Formula: $\text{Variance} = \Sigma(x - \text{mean})^2 / n$ $\text{Standard Deviation} = \sqrt{\text{Variance}}$

Method A: deviations from mean (5.0) = -1, 0, 1; squared = 1, 0, 1; sum = 2

Variance (A) = $2 / 3 = 0.667$ Standard Deviation (A) = $\sqrt{0.667} = 0.816$

Method B: deviations from mean (7.667) = -1.667, 0.333, 1.333; squared = 2.778, 0.111, 1.778; sum = 4.667

Variance (B) = $4.667 / 3 = 1.556$ Standard Deviation (B) = $\sqrt{1.556} = 1.247$

Since Method A has a lower standard deviation (0.816) compared to Method B (1.247), Method A shows less variation between frames and is therefore more consistent and stable in its motion estimates, even though Method B records higher absolute motion values overall.